

Can PCA Recover Hidden Generating Patterns?

Random Matrix Experiments on Spiked Covariance Models

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Motivation

Everything starts from a data-matrix A in shape $T \times N$, T sample-observations on N objects. To spot relations between N objects by sample-observations, we centralize A and compute sample-covariance matrix C :

$$C = \frac{1}{T}A^\top A \quad , A \in \mathbb{R}^{T \times N},$$

By definition, C_{ij} is the sample covariance between object i and object j . But, we are not fulfilled by only spotting sample-covariance between objects, we want more: If the system follows a data-generating process, can we trace back this implicit generation-pattern merely through samples we observe?

To trace implicit-pattern from samples, PCA comes to sight. PCA is an optimization problem: For a covariance matrix Σ , find

$$\max_{\|v\|=1} v^\top \Sigma v.$$

For samples, we have $\Sigma = C = \frac{1}{T}A^\top A$. So:

$$\max_{\|v\|=1} v^\top \frac{1}{T}A^\top A v = \frac{1}{T}\|Av\|^2$$

With $\|v\| = 1$, it exactly shows the variation of current-samples when projected along the direction v . Largest variation direction is "information-intensive", hence it may correlates to the real-generating pattern.

SVD helps finding principal components numerically, it decomposes C as:

$$C = V \frac{S^2}{T} V^\top.$$

Decomposition automatically pops out principal components one by one, showing a correspondence between PC and eigenvalues.

In this project, we research scenario of low-rank factor model as real-DGP for system:

$$A = FL^\top + E.$$

Carry out mathematical reasonings and numerical experiments to answer question above: Can we trace back real-generation pattern only with samples?

Note. Throughout the project, we work in the random matrix regime, which is:

$$N, T \rightarrow \infty, \quad \frac{N}{T} \rightarrow q \in (0, \infty).$$

Pure Noise Benchmark

Let's start with a system of pure-noise outputs:

$$A = E, \quad E_{ti} \sim N(0, \sigma^2).$$

Theorem (Marchenko--Pastur law).¹

Let $X \in \mathbb{R}^{T \times N}$ have i.i.d. entries with mean 0 and variance σ^2 . Under the RMT-regime $N, T \rightarrow \infty$ with $\frac{N}{T} \rightarrow q \in (0, \infty)$, the empirical spectrum of $C_X = \frac{1}{T}X^\top X$ is asymptotically contained in the MP-bulk

$$[\lambda_-, \lambda_+] = [\sigma^2(1 - \sqrt{q})^2, \sigma^2(1 + \sqrt{q})^2].$$

We treat it as a benchmark for research meanwhile verifying MP-law on empirical-spectrum.

Experiment.

Setup:

$T = 3000$, $N = 1000$, n_trials = 50.

Description:

Generating 50 pure-noise data matrix, centralizing them, put all 50 sets of eigenvalues into a frequency-histogram, explicitly showing the fraction of eigenvalues buried in MP-bulk.

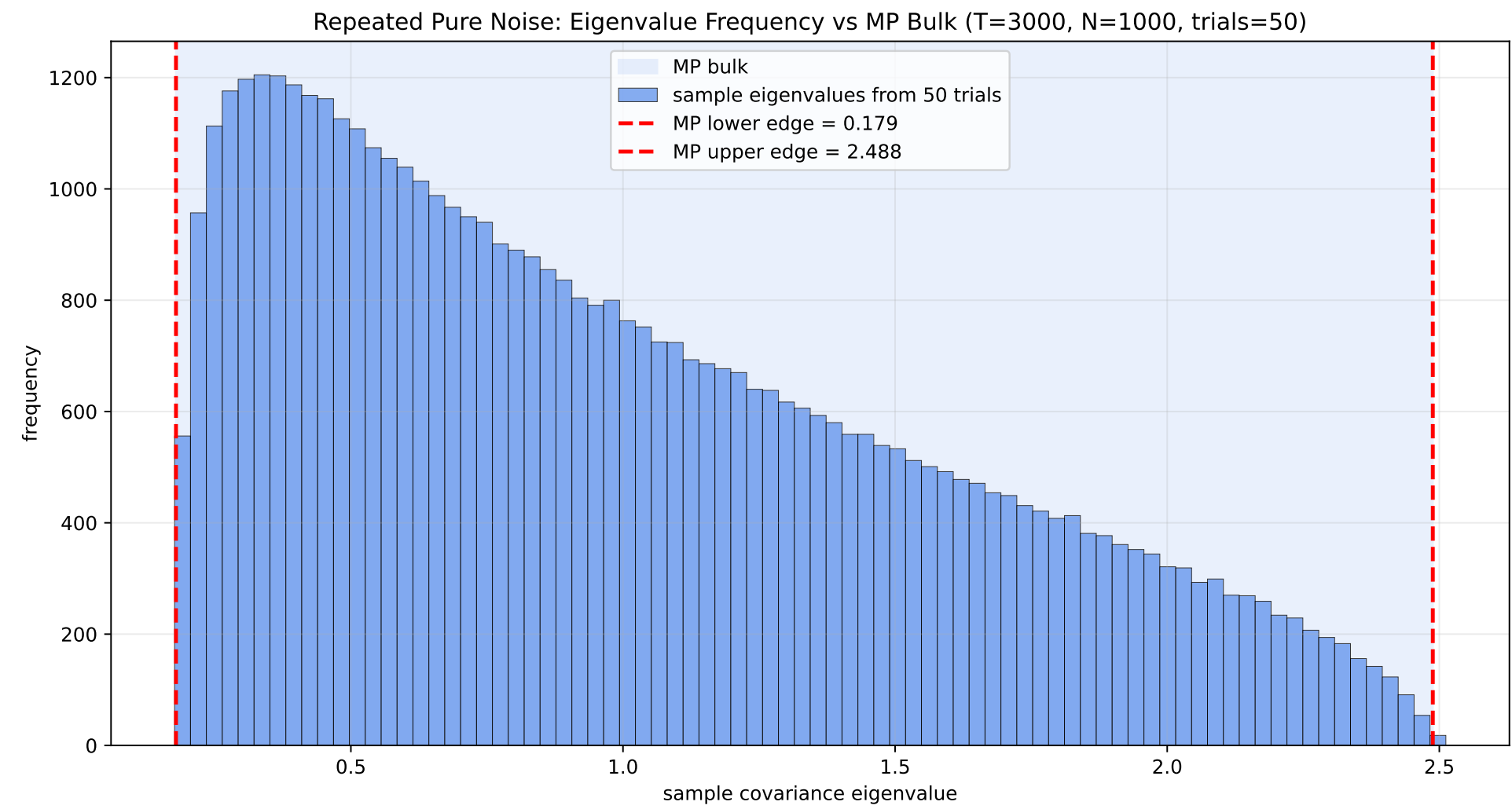


Figure 1: Across 50 pure-noise trials, pooled sample covariance eigenvalues are essentially contained in the MP bulk.

Noise has principal components too.

For pure-noise matrix, PCA still works, but the principal component it returns make no sense, as we know this system simply gives random outputs instead following certain pattern – like factors we will see below.

One-Factor Spike

Now, we assume the system has real-DGP as a one-factor model:

$$a_t = f_l l + e_t.$$

Key Assumptions:

- $f_t \sim N(0, \tau^2)$.
- $l \in \mathbb{R}^{N \times 1}$ is unknown-fixed, with $\|l\| = 1$.
- $e_{ti} \sim N(0, \sigma^2)$.
- f_t and e_t are independent.

By Assumptions above, the population covariance matrix is:

$$\Sigma = \mathbb{E}[a_t a_t^\top] = \sigma^2 I_N + \tau^2 l l^\top.$$

Spots:

- l is Σ 's eigenvector with eigenvalue of $\sigma^2 + \tau^2$.
- Any v orthogonal to l is eigenvector with eigenvalue σ^2 .
- Σ 's eigenvalues are $\tau^2 + \sigma^2, \sigma^2, \dots, \sigma^2$ with one spike.

Question: Is there a spiked eigenvalue in empirical-spectrum?³ If there is, can it recover back the direction of l , since the population spike corresponds to l – the real loading direction in system.

BBP transition

Theorem (BBP phase transition for factor model).²

For finite-rank factor model under RMT-regime, we compare normalized spike strength $\theta_i = \frac{\tau_i^2}{\sigma^2}$ with a threshold $\sqrt{\frac{N}{T}}$. Let

$$\text{card} \left\{ i \mid \theta_i > \sqrt{\frac{N}{T}} \right\} =: k.$$

BBP theory shows, asymptotically, number of outlier eigenvalues of empirical spectrum is exactly k , while the majority of empirical eigenvalues are still buried in MP-bulk.

In one-factor model, we only need to compare θ with $\sqrt{\frac{N}{T}}$. Moreover than estimated-counting, BBP theory also shows: In one-factor model, when $\theta > \sqrt{q}$, PC1 of sample-covariance matrix starts to align with real loading direction l .

Experiment.

Goals:

- asymptotic-count of outlier eigenvalue;
- recovery of loading direction.

Setup:

$T = 600$, $N = 180$, $\sigma = 1$, $\tau = 0, 0.2, \dots, 4.0$, n_trials = 30; crit. $\theta = 0.548$ ($\tau = 0.740$).

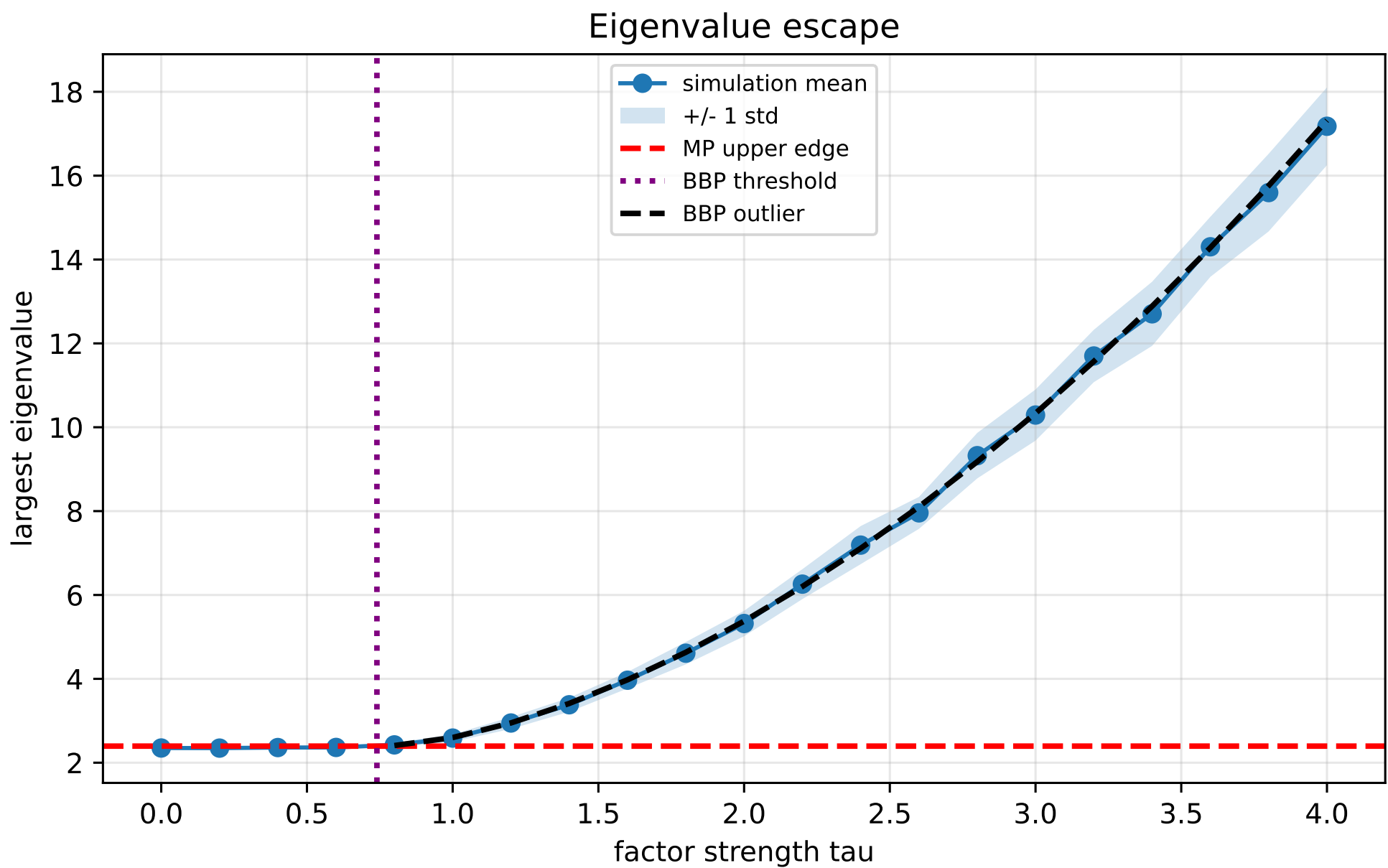


Figure 2: Eigenvalue escape: above threshold, the top eigenvalue separates from the MP bulk.

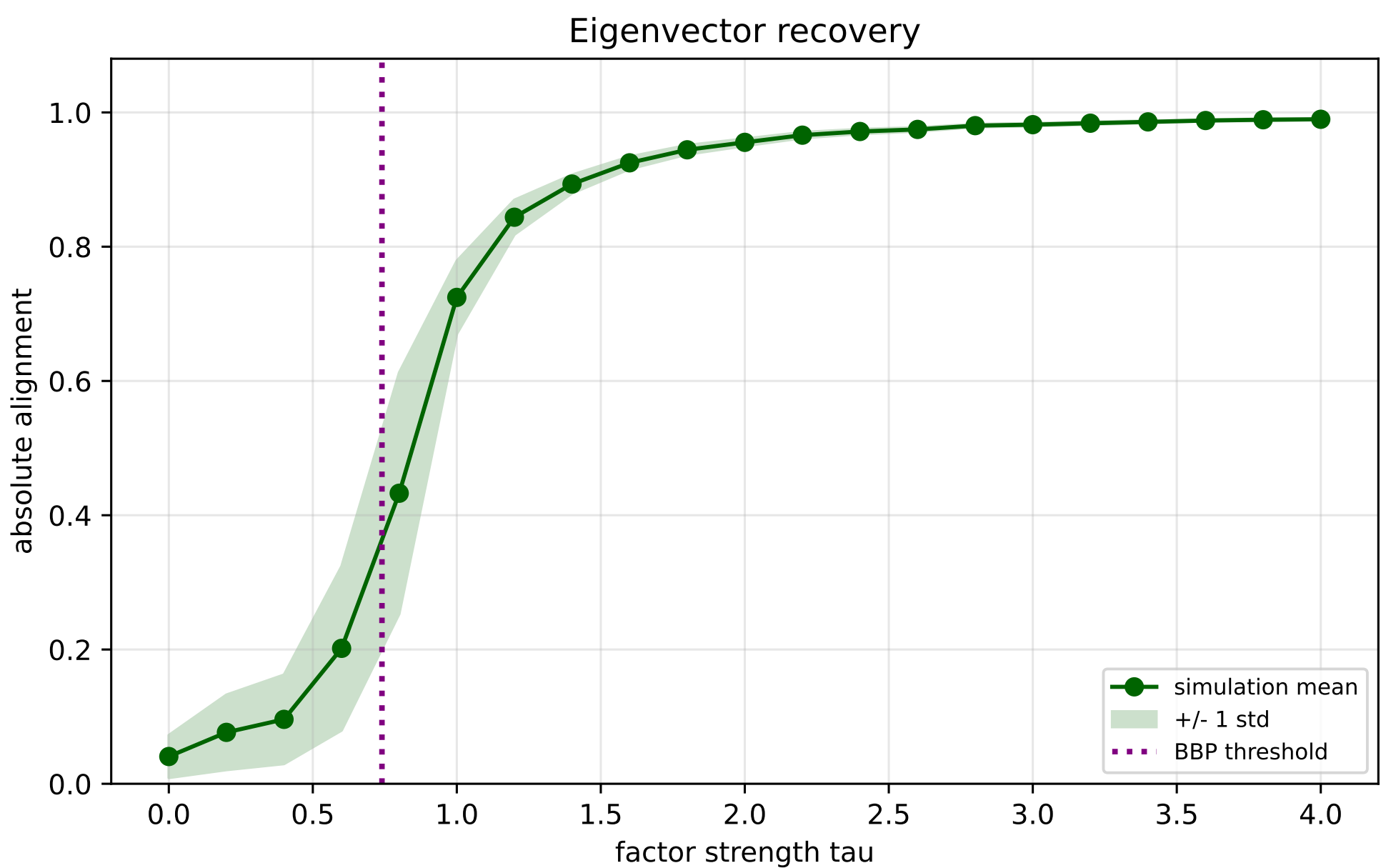


Figure 3: Eigenvector recovery: above threshold, PC1 aligns with the true loading direction.

Multi-Factor Extension

Finally, we extend to k-factor model, i.e. system with real-DGP in form of:

$$a_t = \sum_{i=1}^k f_{ti} l_i + e_t,$$

Key Assumptions:

- $f_{ti} \sim N(0, \tau_i^2)$.
- $l_i \in \mathbb{R}^{N \times 1}$ fixed, with $L^\top L = I_k$.
- $e_{ti} \sim N(0, \sigma^2)$.
- f_{ti} is independent of e_t .

Similar approach to one-factor model, we consider population covariance

matrix:

$$\Sigma = \sigma^2 I_N + L \text{diag}(\tau_1^2, \dots, \tau_k^2) L^\top.$$

Σ has k spiked eigenvalues with different "strengths":

$$\sigma^2 + \tau_1^2, \dots, \sigma^2 + \tau_k^2.$$

We cannot match sample eigenvalues one-to-one with factors. But BBP predicts the asymptotic count of detectable spikes. Regarding threshold stated above:

$$k := \text{card} \left\{ i \mid \theta_i > \sqrt{\frac{N}{T}} \right\}$$

Experiment.

Goals:

- asymptotic count of spikes;
- subspace recovery of detected-spike eigenvalue.

Setup:

$T = 600$, $N = 180$, $k = 3$, $\tau = (1.6, 0.9, 0.5)$, m_detectable = 2, n_trials = 50; crit. $\theta = 0.548$ ($\tau = 0.740$).

Note:

Each of trial, we use real-detected-spike back for subspace alignment experiment.

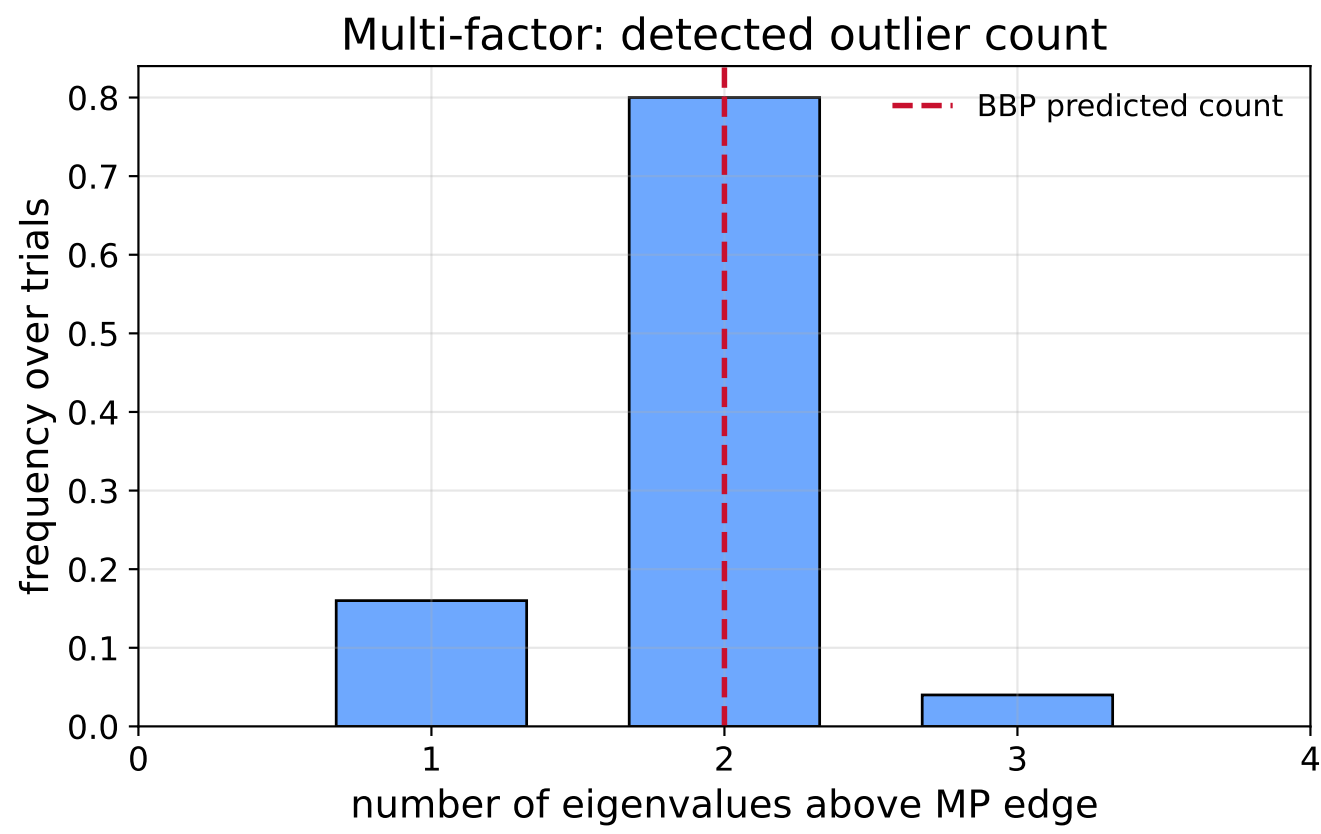


Figure 4: With factor strengths (1.6, 0.9, 0.5), two factors are supercritical. The observed outlier count concentrates near the BBP prediction.

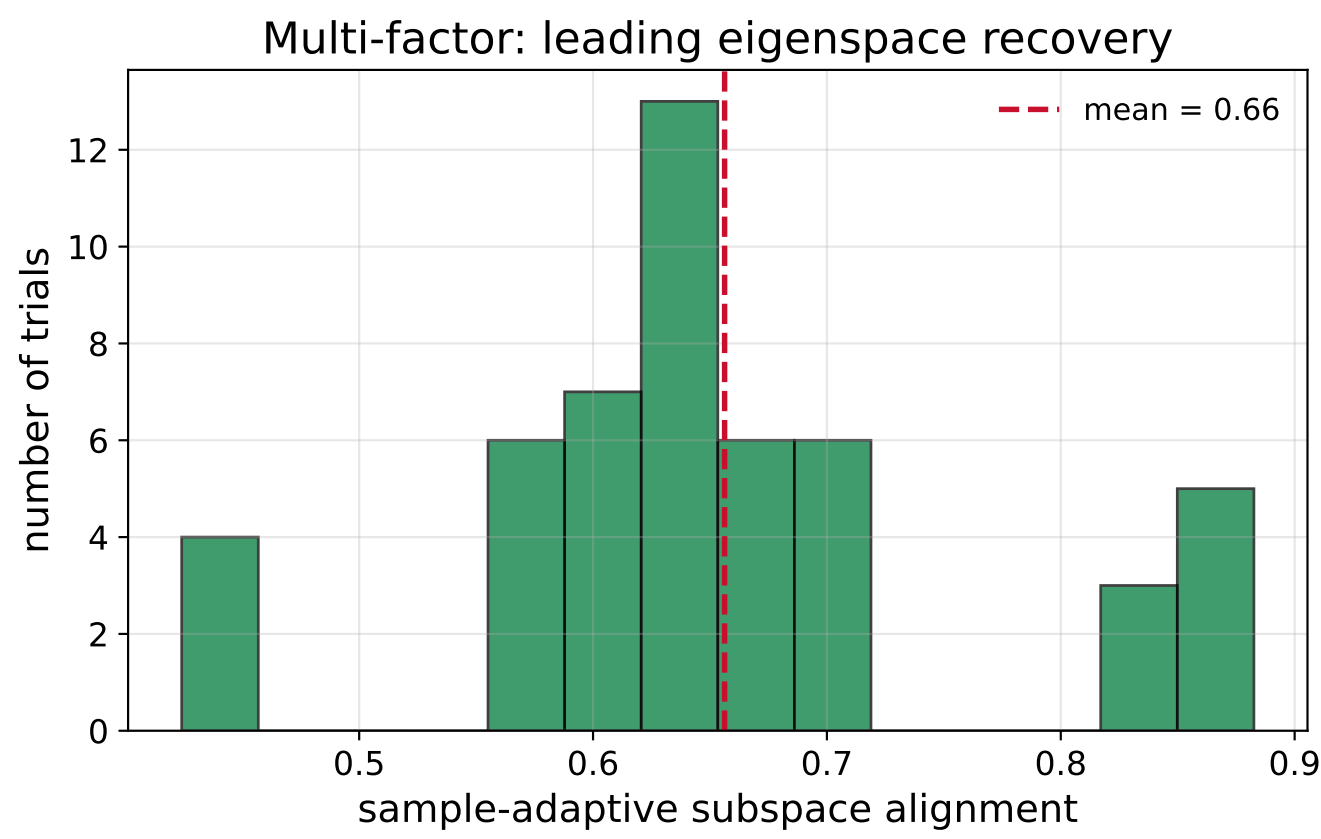


Figure 5: Recovery check: the sample principal subspace aligns with the hidden loading subspace once enough factors are detectable.

PC recovery: $|\langle \hat{v}_1, l_1 \rangle|^2 = 0.852$, $|\langle \hat{v}_2, l_2 \rangle|^2 = 0.375$, $|\langle \hat{v}_3, l_3 \rangle|^2 = 0.016$.

References

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